Mathieu Waharte

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Profile

Software Engineer, specializing in Full-Stack Development and Artificial Intelligence.

Education

Polytech Paris-Saclay (Université Paris-Saclay) — Orsay, France
Diplôme d'Ingénieur (M.Sc.) in Computer Science & Applied Mathematics — *2023 – Present*
- Specialization : Artificial Intelligence, Scientific Computing, Distributed Systems, and Cybersecurity

Paris-Saclay University — Orsay, France
Bachelor of Science (B.S.) in Computer Science - with high honors — *2021 – 2023*
- Supervised Learning, Object-Oriented and Functional Programming, Logic, Computability

L'Essouriau High School, Paris-Saclay University — Les Ulis, France
Intensive Preparatory Class in Math and Physics for Engineering School (CPGE) — *2019 – 2021*
- Mathematics, Physics, Modeling, Problem Solving

Experience

Fullstack Engineer Apprentice — 2023 – Present
Dassault Systèmes — *Vélizy-Villacoublay, France*
- Full-stack design and development for the Scientific Notebook and Reaction Planner applications in an Agile/Scrum environment (Angular, Java, TypeScript, Python, SPARQL) with maintainable documentation.
- Led the development (using design patterns) of a distributed computing queue, from specification and PoC to production, coordinating with Dev/PM/QA/UX teams, resulting in an increase from **40 to 100% reliability, even under load**.
- Engineered a robust **LangChain AI agent** and an **MCP server** to convert molecules into a standardized format, compute synthesizability scores, and interpret them from natural-language prompts through dedicated tools. Validated the system with a mock MCP server, **PACT** flow contracts, and **tests evaluating LLM routing decisions**. All errors are handled through custom states and pre/post-processing.
- **Autonomously** delivered features across **multiple environments** (Java, Node.js, Docker, Python, Angular), ensuring quality through unit/integration testing (Jasmine, Page Objects, PCS) and code reviews. Provided rigorous training and **documentation** for team use.

Research Intern (Natural Language Processing - NLP) — June 2025 – August 2025
Department of Artificial Intelligence, Faculty of ICT, University of Malta — *Malta*
- Conducted a literature review on **Text-to-Speech** (TTS). Developed a paper search application with ranking and filtering via Streamlit, enabling analysis of several thousand articles in mere hours.
- Processed audio data (10h), performed **fine-tuning** and inference on the XTTS model using Google Colab (**PyTorch** & GPT-2) to integrate Maltese without regression. Code is available on GitHub and Hugging Face.

Projects

Band aware dual perspective ResNet for Escampe | *Negamax, ResNet, PyTorch, Java, Machine Learning* | — March 2026
- Engineered **BandDPR**, a custom 730K-parameter **Siamese ResNet** evaluation model featuring dual-perspective CNN encoders, piece-relative spatial mapping (HalfKP), and a direct output bypass for forced-pass detection.
- Implemented adversarial search via **Negamax with Alpha-Beta pruning**, optimized with **Iterative Deepening**, Killer Heuristic, and History Heuristic for aggressive pruning.
- Bootstrapped a dataset of 3 million positions (6M mirrored) using parallel depth-7 minimax search, and trained the model with AdamW and Cosine Annealing with dropout regularization.

Hackathon Deloitte x Google Cloud - 1st Place Winner | *Google Cloud, Agent Development Kit* | — November 2025
- Developed a **distributed multi-agent system** using **ADK** and the **MCP** protocol to automate marketing campaign creation, with specialized agents communicating through the A2A protocol.
- Integrated cloud-native services for demographic queries via BigQuery, multimodal content generation with Gemini and Imagen, and deployment on Cloud Run and Vertex AI Agent Engine.

Dynamic Simulation - Magnus Carlos | *SDL2, C++, Physical Modeling* | — 2024
- **Managed Agile** development of a scientific simulation and 3D rendering for the Magnus effect with a team of 6 students.
- Performed numerical solving and implemented the system in **C++** (real-time visualization with **SDL2**) to reproduce, model, and dynamically predict the behavior of a complex physical system (trajectories, fluids).

Technical Skills

Languages : JavaScript/TypeScript (Angular, Node.js, Express, React, Vue 3), C++, Java, Python (FastAPI), C, SQL, Bash
Quality & Software Testing : PACT, Jasmine, Jest, Karma, Page Objects, PCS (JMeter), JUnit, Mocks, TDD
Artificial Intelligence & Data : LLM, LoRA, LangChain, MCP, RAG, PyTorch, NumPy, pandas, OpenCV, scikit-learn
Machine Learning : AI agents, model training and evaluation, data preprocessing, Big Data, PCA, un/supervised learning
Simulation & Numerical Computation : ODE/PDE solving, probabilistic models, MPI, OpenMP, CUDA
Design, Methods & Databases : API REST, Design Patterns, UML, Postgres, MySQL, RDF, SQLite, Prisma, Redis
Tools and DevOps : Git, CI/CD, Docker, Postman, OpenAPI, Packaging, Tomcat, VMware, Hugging Face, LaTeX

Certificates and Languages

Model Context Protocol : Advanced Topics - Anthropic
Label Handimanagement - Companieros & Dassault Systèmes
MOOC SecNumAc cybersecurity - ANSSI
English : C1 (TOEIC, 990), full professional proficiency